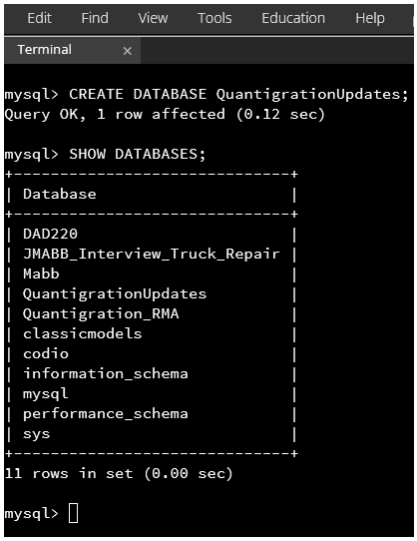
Database Architecture and Integrity

Overview

This Project serves as the major update to the QuantigrationRMA database build. The database was overhauled and renamed QuantigrationUpdates. A company initiative to rename customers to collaborators was incorporated. Special attention was paid to the data type field length to ensure no data loss occurred when importing the client's data. Additionally, manually inserted customer and order records were compared against prior client hard copies received, and any adjustments were made for the client's update.

Step One: Create a Database

1. A new database titled QuantigrationUpdates was created.
    a. CREATE DATABASE and SHOW DATABASES were used to create and verify the creation.
    b. The screenshot below includes the MySQL commands used.



2. The tables for Customers, Orders, and RMA were created using the client-provided ERD.
    a. CREATE TABLE, SHOW TABLES, and DESCRIBE Customers were used to create and verify the creation.

b. The screenshot below includes the MySQL commands used.

```
Edit  Find  View  Tools  Education  Help  Configur.
Terminal

mysql> USE QuantigrationUpdates;
Database changed
mysql> CREATE TABLE Customers (
    -> CustomerID INT,
    -> FirstName VARCHAR(25),
    -> LastName VARCHAR(25),
    -> Street VARCHAR(50),
    -> City VARCHAR(50),
    -> State VARCHAR(25),
    -> ZipCode VARCHAR(10),
    -> Telephone VARCHAR(15),
    -> PRIMARY KEY (CustomerID)
    -> );
Query OK, 0 rows affected (0.23 sec)

mysql> SHOW TABLES;
+--------------------------------+
| Tables_in_QuantigrationUpdates |
+--------------------------------+
| Customers                      |
+--------------------------------+
1 row in set (0.00 sec)

mysql> DESCRIBE Customers;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| CustomerID | int         | NO   | PRI | NULL    |       |
| FirstName  | varchar(25) | YES  |     | NULL    |       |
| LastName   | varchar(25) | YES  |     | NULL    |       |
| Street     | varchar(50) | YES  |     | NULL    |       |
| City       | varchar(50) | YES  |     | NULL    |       |
| State      | varchar(25) | YES  |     | NULL    |       |
| ZipCode    | varchar(10) | YES  |     | NULL    |       |
| Telephone  | varchar(15) | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
```

a. CREATE TABLE, SHOW TABLES, and DESCRIBE Orders were used to create and verify the creation. Note the use of Foreign Key CustomerID for the Orders table.

b. The screenshot below includes the MySQL commands used.

```
Edit  Find  View  Tools  Education  Help  Configur.  Project
Terminal

mysql> CREATE TABLE Orders (
    -> OrderID INT,
    -> CustomerID INT,
    -> SKU VARCHAR(20),
    -> Description VARCHAR(75),
    -> PRIMARY KEY (OrderID),
    -> FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
    -> );
Query OK, 0 rows affected (0.31 sec)

mysql> SHOW TABLES;
+--------------------------------+
| Tables_in_QuantigrationUpdates |
+--------------------------------+
| Customers                      |
| Orders                         |
+--------------------------------+
2 rows in set (0.00 sec)

mysql> DESCRIBE Orders;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| OrderID     | int         | NO   | PRI | NULL    |       |
| CustomerID  | int         | YES  | MUL | NULL    |       |
| SKU         | varchar(20) | YES  |     | NULL    |       |
| Description | varchar(75) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

a. CREATE TABLE, SHOW TABLES, and DESCRIBE RMA were used to create and verify the creation. Note the use of Foreign Key OrderID for the RMA table.

b. The screenshot below includes the MySQL commands used.

```
Edit   Find   View   Tools   Education   Help   Configur.
Terminal

mysql> CREATE TABLE RMA (
    -> RMAID INT,
    -> OrderID INT,
    -> Step VARCHAR(50),
    -> Status VARCHAR(15),
    -> Reason VARCHAR(15),
    -> PRIMARY KEY (RMAID),
    -> FOREIGN KEY (OrderID) REFERENCES Orders(OrderID)
    -> );
Query OK, 0 rows affected (0.45 sec)

mysql> SHOW TABLES;
+--------------------------------+
| Tables_in_QuantigrationUpdates |
+--------------------------------+
| Customers                      |
| Orders                         |
| RMA                            |
+--------------------------------+
3 rows in set (0.00 sec)

mysql> DESCRIBE RMA;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| RMAID   | int         | NO   | PRI | NULL    |       |
| OrderID | int         | YES  | MUL | NULL    |       |
| Step    | varchar(50) | YES  |     | NULL    |       |
| Status  | varchar(15) | YES  |     | NULL    |       |
| Reason  | varchar(15) | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

Step Two: Load and Query the Data

1. The client provided CSV files for Customers, Orders, and RMA, which were loaded into the QuantigrationUpdates database.
2. The screenshot below shows the import of the 37,994 records.

```
Edit  Find  View  Tools  Education  Help  Configur.  Project I..  Configur.
Terminal
+---------+-------------+------+-----+---------+-------+
5 rows in set (0.01 sec)

mysql> LOAD DATA INFILE '/home/codio/workspace/customers.csv' INTO TABLE Customers
    -> FIELDS TERMINATED BY ','
    -> LINES TERMINATED BY '\r\n'
    -> ;
Query OK, 37994 rows affected (2.38 sec)
```

```
Edit  Find  View  Tools  Education  Help  Configur.  Project I..  Configur
Terminal
mysql> LOAD DATA INFILE '/home/codio/workspace/orders.csv' INTO TABLE Orders
    -> FIELDS TERMINATED BY ','
    -> LINES TERMINATED BY '\r\n'
    -> ;
Query OK, 37994 rows affected (3.28 sec)
Records: 37994  Deleted: 0  Skipped: 0  Warnings: 0

mysql>
```

```
Edit  Find  View  Tools  Education  Help  Configur.  Project I..
Terminal
mysql> LOAD DATA INFILE '/home/codio/workspace/rma.csv' INTO TABLE RMA
    -> FIELDS TERMINATED BY ','
    -> LINES TERMINATED BY '\r\n'
    -> ;
Query OK, 38162 rows affected (3.53 sec)
Records: 38162  Deleted: 0  Skipped: 0  Warnings: 0

mysql>
```

a. The MySQL commands are recaptured here in my backup Notepad++ file for checking my code.

```
// Load CSV files

LOAD DATA INFILE '/home/codio/workspace/customers.csv' INTO TABLE Customers
FIELDS TERMINATED BY ','
LINES TERMINATED BY '\r\n'
;

LOAD DATA INFILE '/home/codio/workspace/orders.csv' INTO TABLE Orders
FIELDS TERMINATED BY ','
LINES TERMINATED BY '\r\n'
;

LOAD DATA INFILE '/home/codio/workspace/rma.csv' INTO TABLE RMA
FIELDS TERMINATED BY ','
LINES TERMINATED BY '\r\n'
;
```

3. Basic queries against the imported tables to demonstrate the validity of the data import and provide some targeted analysis.

a. Query of the customer orders from customers located in Framingham, Massachusetts.

i. The query requires a JOIN between the Customers and Orders tables. There are 505 records.

ii. The screenshot of the query and the results are below.

```
mysql> SELECT COUNT(OrderID)
    -> FROM Customers
    -> INNER JOIN Orders
    -> ON Orders.CustomerID = Customers.CustomerID
    -> WHERE Customers.City = 'Framingham' AND Customers.State = 'Massachusetts'
    -> ;
+----------------+
| COUNT(OrderID) |
+----------------+
|            505 |
+----------------+
1 row in set (0.03 sec)
```

b. Query of all the customers located in Massachusetts.

   i. This query does not require a join. There are 982 total customers in the state of Massachusetts. These are customers, not orders.

   ii. The screenshot of the query and the results are below.

```
mysql> SELECT COUNT(CustomerID)
    -> FROM Customers
    -> WHERE Customers.State = 'Massachusetts'
    -> ;
+-------------------+
| COUNT(CustomerID) |
+-------------------+
|               982 |
+-------------------+
1 row in set (0.02 sec)

mysql>
```

4. Manual insertion of records.

   a. Insert four new customers into the Customers table with the information provided by the client. Special attention was paid to the client's address change corrections.

      i. The screenshot below shows the MySQL commands and successful insertion.

```
mysql> INSERT INTO Customers
    -> VALUES
    -> (100004, 'Luke', 'Skywalker', '15 Maiden Lane', 'New York', 'NY', '10222', '212-555-1234'),
    -> (100005, 'Winston', 'Smith', '123 Sycamore Street', 'Greensboro', 'NC', '27401', '919-555-6623'),
    -> (100006, 'MaryAnne', 'Jenkins', '1 Coconut Way', 'Jupiter', 'FL', '33458', '321-555-8907'),
    -> (100007, 'Janet', 'Williams', '55 Redondo Beach Blvd', 'Torrence', 'CA', '90501', '310-555-5678')
    -> ;
Query OK, 4 rows affected (0.06 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
```

b. However, the new records' data for the State field uses abbreviations for states. The data in the Customers.csv file spells out the states; therefore, updates to the customer records were needed.

   i. The screenshot below shows the update to the State field for the new customer records.

```
Edit  Find  View  Tools  Education  Help
Terminal

mysql> UPDATE Customers
    -> SET State = 'New York'
    -> WHERE CustomerID = 100004;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Customers
    -> SET State = 'North Carolina'
    -> WHERE CustomerID = 100005;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Customers
    -> SET State = 'Florida'
    -> WHERE CustomerID = 100006;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Customers
    -> SET State = 'California'
    -> WHERE CustomerID = 100007;
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
```

c. Insert four new orders into the Orders table with the information provided by the client. The data for this update was manually verified before coding.

   i. The screenshot below shows the MySQL commands and successful insertion.

```
Edit  Find  View  Tools  Education  Help  Configure....  Project Index (static)  Configure....  JMABB
Terminal

mysql> INSERT INTO Orders
    -> VALUES
    -> (1204305, 100004, 'ADV-24-10C', 'Advanced Switch 10GigE Copper 24 port'),
    -> (1204306, 100005, 'ADV-48-10F', 'Advanced Switch 10GigE Copper/Fiber 44 port copper 4 port fiber'),
    -> (1204307, 100006, 'ENT-24-10F', 'Enterprise Switch 10GigE SFP+ 24 Port'),
    -> (1204308, 100007, 'ENT-48-10F', 'Enterprise Switch 10GigE SFP+ 48 port')
    -> ;
Query OK, 4 rows affected (0.05 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
```

5. Query Customer table to identify the number of customers in Woonsocket, Rhode Island.

   a. There are 7 customers in Woonsocket, Rhode Island. The screenshot below shows the MySQL commands and results.

```
mysql> SELECT COUNT(CustomerID)
    -> FROM Customers
    -> WHERE Customers.City = 'Woonsocket' AND Customers.State = 'Rhode Island'
    -> ;
+-------------------+
| COUNT(CustomerID) |
+-------------------+
|                 7 |
+-------------------+
1 row in set (0.03 sec)

mysql>
```

6. Update RMA database.

   a. For OrderID value of "5175", query the status and step.

      i. The status and step of OrderID 5175 are ‘Pending’, and ‘Awaiting customer Documentation’. The screenshot below shows the MySQL commands and results.

```
mysql> SELECT OrderID, Status, Step
    -> FROM RMA
    -> WHERE OrderID = 5175
    -> ;
+---------+---------+--------------------------------+
| OrderID | Status  | Step                           |
+---------+---------+--------------------------------+
|    5175 | Pending | Awaiting customer Documentation |
+---------+---------+--------------------------------+
1 row in set (0.00 sec)

mysql>
```

   b. Update the status and step for the OrderID, 5175 to status = "Complete" and step = "Credit Customer Account".

      i. The updated status and step of OrderID 5175 are ‘Complete’, and ‘Credit Customer Account’. The screenshot below shows the MySQL commands and results.

```
Edit  Find  View  Tools  Education  Help  Configure....  Project Index (static)  Configure....
Terminal

mysql> UPDATE RMA
    -> SET Status = REPLACE(Status, 'Pending', 'Complete')
    -> WHERE OrderID = 5175
    -> ;
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE RMA
    -> SET Step = REPLACE(Step, 'Awaiting customer Documentation', 'Credit Customer Account')
    -> WHERE OrderID = 5175
    -> ;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT OrderID, Status, Step
    -> FROM RMA
    -> WHERE OrderID = 5175
    -> ;
+---------+----------+------------------------+
| OrderID | Status   | Step                   |
+---------+----------+------------------------+
|    5175 | Complete | Credit Customer Account |
+---------+----------+------------------------+
1 row in set (0.00 sec)
```

c. Delete RMA records.

i. Before deleting many records for a database, it was essential to determine how many records there were and create a backup output file.

ii. Below are the MySQL commands to determine the number of records with the 'Rejected' reason and to create the backup file.

```
Edit  Find  View  Tools  Education  Help  Configure....  Project Index (static)  Configure....
Terminal
|  1495 |   94392 | Product replacement or account refund processed | Complete | Rejected |
|  1496 |   30140 | Product replacement or account refund processed | Complete | Rejected |
|  1499 |   43471 | Product replacement or account refund processed | Complete | Rejected |
+-------+---------+-------------------------------------------------+----------+----------+
596 rows in set (0.03 sec)

mysql> SELECT *
    -> FROM RMA
    -> WHERE Reason = 'Rejected'
    -> INTO outfile'/home/codio/workspace/RejectedRMA2024-12-04.csv'
    -> FIELDS TERMINATED BY ','
    -> LINES TERMINATED BY '\r\n'
    -> ;
Query OK, 596 rows affected (0.03 sec)

mysql>
```

iii. 596 records were deleted from the RMA table. Below is the screenshot showing the deletion and verification.

```
mysql> DELETE FROM RMA
    -> WHERE Reason = 'Rejected'
    -> ;
Query OK, 596 rows affected (0.13 sec)

mysql> SELECT *
    -> FROM RMA
    -> WHERE Reason = 'Rejected'
    -> ;
Empty set (0.02 sec)

mysql>
```

Step Three: Collaborator Initiative

1. Quantigration has a new initiative to update the term Customer to Collaborator. All occurrences of the word customer need to be updated.
    a. Before a major initiative change, a backup output file needed to be created for any tables being updated. Below are the MySQL commands and results for creating backups of the Customers and Orders tables.

```
mysql> SELECT *
    -> FROM Customers
    -> INTO outfile'/home/codio/workspace/Customers2024-12-04.csv'
    -> FIELDS TERMINATED BY ','
    -> LINES TERMINATED BY '\r\n'
    -> ;
Query OK, 37998 rows affected (0.13 sec)

mysql> SELECT *
    -> FROM Orders
    -> INTO outfile'/home/codio/workspace/Orders2024-12-04.csv'
    -> FIELDS TERMINATED BY ','
    -> LINES TERMINATED BY '\r\n'
    -> ;
Query OK, 37998 rows affected (0.06 sec)
```

    b. Rename all instances of "Customer" to "Collaborator".
        i. The Customers table was renamed to Collaborators. Below are the MySQL commands and verification of the table name change.

```
Edit   Find   View   Tools   Education
Terminal
mysql> RENAME TABLE
    -> Customers TO Collaborators
    -> ;
Query OK, 0 rows affected (0.17 sec)

mysql> SHOW TABLES;
+--------------------------------+
| Tables_in_QuantigrationUpdates |
+--------------------------------+
| Collaborators                  |
| Orders                         |
| RMA                            |
+--------------------------------+
3 rows in set (0.00 sec)

mysql>
```

ii. CustomerID was renamed CollaboratorID for the new Collaborators table.

Below are the MySQL commands and verification of the field name change.

```
Edit   Find   View   Tools   Education   Help   Configure....   Proj
Terminal
mysql> DESCRIBE Collaborators;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| CustomerID | int         | NO   | PRI | NULL    |       |
| FirstName  | varchar(25) | YES  |     | NULL    |       |
| LastName   | varchar(25) | YES  |     | NULL    |       |
| Street     | varchar(50) | YES  |     | NULL    |       |
| City       | varchar(50) | YES  |     | NULL    |       |
| State      | varchar(25) | YES  |     | NULL    |       |
| ZipCode    | varchar(10) | YES  |     | NULL    |       |
| Telephone  | varchar(15) | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
8 rows in set (0.00 sec)

mysql> ALTER TABLE Collaborators
    -> RENAME COLUMN CustomerID to CollaboratorID
    -> ;
Query OK, 0 rows affected (0.15 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESCRIBE Collaborators;
+----------------+-------------+------+-----+---------+-------+
| Field          | Type        | Null | Key | Default | Extra |
+----------------+-------------+------+-----+---------+-------+
| CollaboratorID | int         | NO   | PRI | NULL    |       |
| FirstName      | varchar(25) | YES  |     | NULL    |       |
| LastName       | varchar(25) | YES  |     | NULL    |       |
| Street         | varchar(50) | YES  |     | NULL    |       |
| City           | varchar(50) | YES  |     | NULL    |       |
| State          | varchar(25) | YES  |     | NULL    |       |
| ZipCode        | varchar(10) | YES  |     | NULL    |       |
| Telephone      | varchar(15) | YES  |     | NULL    |       |
+----------------+-------------+------+-----+---------+-------+
8 rows in set (0.00 sec)
```

iii. CustomerID was renamed CollaboratorID for the Orders table due to the foreign key. Below are the MySQL commands and verification of the field name change.

```
mysql> DESCRIBE Orders;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| OrderID     | int         | NO   | PRI | NULL    |       |
| CustomerID  | int         | YES  | MUL | NULL    |       |
| SKU         | varchar(20) | YES  |     | NULL    |       |
| Description | varchar(75) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql> ALTER TABLE Orders
    -> RENAME COLUMN CustomerID to CollaboratorID
    -> ;
Query OK, 0 rows affected (0.22 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESCRIBE Orders;
+----------------+-------------+------+-----+---------+-------+
| Field          | Type        | Null | Key | Default | Extra |
+----------------+-------------+------+-----+---------+-------+
| OrderID        | int         | NO   | PRI | NULL    |       |
| CollaboratorID | int         | YES  | MUL | NULL    |       |
| SKU            | varchar(20) | YES  |     | NULL    |       |
| Description    | varchar(75) | YES  |     | NULL    |       |
+----------------+-------------+------+-----+---------+-------+
4 rows in set (0.01 sec)

mysql>
```

Step Four: Wrapping Up

1. Create an output file of the current state of the Orders table and send the output to a file with a CSV extension.

    a. Below is the screenshot showing the MySQL commands and the results of the output.

```
mysql> SELECT *
    -> FROM Orders
    -> INTO outfile'/home/codio/workspace/Orders2024-12-05.csv'
    -> FIELDS TERMINATED BY ','
    -> LINES TERMINATED BY '\r\n'
    -> ;
Query OK, 37998 rows affected (0.08 sec)

mysql>
```